

South Asia's Digital Revolution

UPI, India Stack, and the DPI Playbook for the Global South

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Executive Summary

India's digital payments infrastructure has reached a scale that is attracting global integration partners and large-scale private capital. The Reserve Bank of India and the European Central Bank have agreed to begin interlinking their domestic payment systems CNA[2025-11-21], and India's Unified Payments Interface has been incorporated into Project Nexus — a cross-border payments network linking Singapore's PayNow, Malaysia's DuitNow, Thailand's PromptPay, the Philippines' InstaPay, and India's UPI CNA[2025-01-19]. In parallel, private capital commitment to India's AI and data infrastructure has reached extraordinary scale: Adani Enterprises announced a US\$100 billion investment to build renewable energy-powered AI-ready data centres by 2035 CNA[2026-02-17], and a Reliance Industries joint venture committed \$11 billion over five years to develop 1 gigawatt of AI data capacity in Andhra Pradesh CNA[2025-11-26].

India's fintech lending sector is expanding rapidly but generating new risks. The loan portfolio of fintech-driven non-banking financial companies hit 2.1 trillion rupees as active loans grew by 25.6 per cent year on year CNA[2026-08-01]. Borrowing pressure on young Indians is rising in tandem, with reports of financial stress among urban users of digital lending apps CNA[2026-08-01].

Part I — UPI: Payments Infrastructure Going Global

Cross-Border Linkages

Singapore's PayNow and India's UPI were linked by July 2022, enabling instant, low-cost fund transfers directly between bank accounts in the two countries CNA[2021-09-14]. This bilateral corridor was subsequently embedded in the broader Project Nexus architecture, under which the central banks of Singapore, Malaysia, Thailand, the Philippines, and India are collaborating to create a global network for instant cross-border payments CNA[2025-01-19]. Project Nexus represents ASEAN taking the lead in payment system interoperability, with UPI serving as India's node in the network CNA[2025-01-19].

At the continental level, the Reserve Bank of India and the European Central Bank agreed in November 2025 to start the initial phase of interlinking their domestic payment systems CNA[2025-11-21]. The direction of integration — UPI connecting outward to ASEAN payment rails and inward toward European systems — reflects the network effects that accrue to payment infrastructure that achieves sufficient domestic scale.

Biometric Authentication Layer

India announced in October 2025 that it would allow UPI users to approve payments using facial recognition and fingerprints, rolling out biometric authentication for the payments network

CNA[2025-10-07]. This extends the identity verification layer that underpins UPI's security architecture into a new authentication modality, building on India's existing biometric identity infrastructure.

Part II — Digital Financial Inclusion: Fintech Lending and Its Risks

The Fintech Lending Boom

Fintech lending — digital lending on platforms using technology to assess credit and issue loans — has grown rapidly in India. By mid-2025, the loan portfolio of fintech-driven non-banking financial companies had reached 2.1 trillion rupees, with the number of active loans growing 25.6 per cent year on year, according to credit bureau CRIF High Mark CNA[2026-08-01]. The ease and speed of digital credit access have driven borrowing among younger urban Indians who previously lacked access to formal bank credit CNA[2026-08-01].

The same accessibility that drives financial inclusion is generating stress. Borrowers report difficulty managing repayment timelines across multiple lending app loans, and the psychological pressure of digital debt collection is a documented feature of the ecosystem CNA[2026-08-01]. The regulatory question — how to preserve the credit access gains while addressing predatory lending behaviour — remains live.

Amazon's Entry into Direct Lending

Amazon completed its acquisition of Axio in September 2025, securing a direct lending licence in India through the acquisition CNA[2025-09-04]. The entry of a global e-commerce platform into direct lending marks a further deepening of the fintech-commerce integration in India's digital economy, with implications for how credit is bundled with commerce for Indian consumers.

Part III — Data Infrastructure: The Capital Commitment Wave

Adani and Reliance: India's Hyperscale Bet

Two of India's largest conglomerates have made commitments to AI-ready data infrastructure at a scale that would, if executed, reshape India's position in the global AI supply chain. Adani Enterprises announced in February 2026 that it will invest US\$100 billion to build renewable energy-powered AI-ready data centres by 2035, targeting the creation of the world's largest integrated data centre platform CNA[2026-02-17]. A Reliance Industries joint venture separately committed \$11 billion over five years to develop 1 gigawatt of AI data capacity in the southern Indian state of Andhra Pradesh CNA[2025-11-26].

These commitments reflect both the global surge in demand for AI compute infrastructure and India's ambition to capture a portion of that infrastructure buildout domestically rather than remain dependent on hyperscale capacity hosted outside the country.

Part IV — AI Adoption: Structural Constraints

The Cloud Infrastructure Gap

A structural constraint on India's AI adoption has been identified in research from Carnegie Endowment: the absence of a large native install base of on-demand cloud-computing infrastructure in India puts many recent advances in AI out of the reach of government-funded research labs. Additionally, many Indian industries cannot risk storing their data outside of India, yet lack control over the algorithms that process it when accessed via foreign cloud services [RAG: SCI_TECH_RAG — Carnegie: India And The Artificial Intelligence Revolution (2016-01-01)]. The large-scale domestic data centre investments

announced by Adani and Reliance in 2025–2026 are directly relevant to this structural gap.

Conclusion

South Asia's digital infrastructure story in 2025–2026 is one of external integration and internal capital mobilisation. UPI has moved beyond India's borders into ASEAN's Project Nexus and toward ECB linkage CNA[2025-01-19], CNA[2025-11-21]. Biometric authentication has extended UPI's security layer CNA[2025-10-07]. Private capital at the scale of US\$100 billion (Adani) and \$11 billion (Reliance) is being committed to AI-ready data infrastructure CNA[2026-02-17], CNA[2025-11-26] — addressing the structural absence of domestic compute capacity that has historically constrained India's AI ecosystem [RAG: SCI_TECH_RAG — Carnegie: India And The Artificial Intelligence Revolution (2016-01-01)]. The fintech lending sector has reached 2.1 trillion rupees in active loan portfolio CNA[2026-08-01], creating both inclusion gains and debt stress risks that regulatory frameworks have yet to fully address.

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